

Go Back-N (GBN) Protocol: Technical Analysis

Summary

Go Back-N (GBN) is a sliding window flow control protocol designed to improve efficiency over Stop-and-Wait by allowing multiple packets to be transmitted before receiving acknowledgements (ACKs).

Its defining characteristics are:

- **Sender window size (W_s) = N**
- **Receiver window size (W_r) = 1**
- **Cumulative acknowledgements**
- **Retransmission of entire window after a loss ("Go Back-N")**

Because the receiver only accepts packets in strict order, any out-of-order packet is discarded. If a packet is lost, the sender retransmits that packet and all subsequent packets in the current window.

1. Overview of Sliding Window Protocols

The sliding window mechanism improves link utilization compared to Stop-and-Wait.

Mechanism

- The sender transmits a window of **N** packets without waiting for individual ACKs.
- As ACKs arrive, the window slides forward.
- This keeps the channel busy during round-trip time (RTT).

Protocol Comparison

Feature	Stop-and-Wait	Go Back-N (GBN)	Selective Repeat (SR)
Sender Window (W_s)	1	N	N
Receiver Window (W_r)	1	1	N
Out-of-Order Packets	Not applicable	Discarded	Buffered

GBN increases efficiency by allowing pipelined transmission while keeping the receiver logic simple.

2. Technical Specifications of GBN

GBN is commonly represented as **GBN-N**, where N is the sender window size.

Window Sizes

- $W_s = N$
- $W_r = 1$ (fixed)

To outperform Stop-and-Wait:

$$N > 1$$

Operational Logic

- The sender can transmit up to N packets without waiting.
- ACK arrival slides the window forward.
- The receiver only accepts the next expected packet.
- Any packet received out-of-order is discarded.

This strict ordering simplifies receiver design but increases retransmission cost during loss.

3. Error Handling and Retransmission

Retransmission Process

1. **Packet Loss Occurs**
Suppose Packet 3 is lost.
2. **Receiver Behavior**
 - Receiver continues expecting Packet 3.
 - Packets 4, 5, 6, etc., even if received correctly, are discarded.
3. **Timer Mechanism**
 - Sender maintains a timer for only the **oldest unacknowledged packet**.
4. **Timeout and Go Back-N Action**
 - When timeout occurs, sender retransmits:
 - The lost packet
 - All subsequent packets in the current window

This results in potentially redundant retransmissions.

4. Timer Management

Unlike Selective Repeat:

- GBN uses **only one timer**.
- The timer is associated with the **oldest unacknowledged frame**.
- If this timer expires, the sender retransmits the entire window starting from that frame.

This reduces implementation complexity compared to maintaining multiple timers.

5. Acknowledgement Mechanism

GBN primarily uses **cumulative acknowledgements**.

Independent ACK

Each packet acknowledged separately.

Cumulative ACK

ACK with number n implies:

All packets up to $n-1$ are successfully received, and the receiver now expects packet n .

Timer Constraint

To avoid unnecessary retransmissions:

$$T_{ACK} < T_{out}$$

The receiver must send ACKs before the sender timeout expires.

6. Sequence Number and Window Size Relationship

To avoid ambiguity between new and old packets:

$W_s + W_r \leq \text{Available Sequence Numbers}$

For GBN:

Since $W_r = 1$,

$W_s \leq \text{Total Sequence Numbers} - 1$

K-bit Sequence Numbers

If k bits are used:

Total sequence numbers = 2^k

Maximum sender window size:

$W_s(\text{max}) = 2^k - 1$

$W_r = 1$

This ensures no duplicate packet confusion.

7. Quantitative Transmission Analysis

Scenario A: GBN-3, 10 Packets, Every 5th Packet Lost

Window size = 3

Loss pattern: 5th transmission lost

When Packet 5 is lost:

- Packets 6 and 7 are already transmitted.
- Receiver discards them.
- Sender retransmits Packets 5, 6, and 7.

Total transmissions required: 18

Scenario B: GBN-4, 10 Packets, Every 6th Packet Lost

Window size = 4

When Packet 6 is lost:

- Sender retransmits Packets 6, 7, 8, 9.

Total transmissions required: 17

Scenario C: GBN-3, 10 Packets, Every 4th Packet Lost

Window size = 3

Loss frequency equals or is less than window size.

Result:

- Frequent timeouts
- Repeated retransmission of same window
- Rapid growth in total transmissions

Total transmissions required: 27

This demonstrates how GBN performance degrades severely under certain periodic loss patterns.

8. Key Conclusions

Receiver Simplicity

$W_r = 1$ minimizes buffering and implementation complexity.

In-Order Delivery Guarantee

Upper layers always receive packets in correct sequence.

Retransmission Overhead

Main drawback: retransmission of packets that may have arrived correctly but were discarded.

Performance Sensitivity

Efficiency strongly depends on:

- Window size
- RTT
- Loss probability

Evolution to Selective Repeat

The inefficiency of discarding out-of-order packets led to Selective Repeat (SR), where:

- $W_r > 1$
- Only lost packets are retransmitted

